

BUKGANGNEUNG, Korea, Republic of

WMO: 471040

Lat: 37.8044N

Lon: 128.8553E

Elev: 263

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.6	(c) 19.2	(d) -14.3	(e) 2.6	(f) 22.2	(g) -10.5	(h) 3.2	(i) 23.6	(j) 16.5	(k) 29.4	(l) 14.9	(m) 31.7	(n) 6.7	(o) 160	(p) 0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.1	(c) 89.9	(d) 74.1	(e) 87.5	(f) 73.3	(g) 84.7	(h) 71.7	(i) 77.3	(j) 84.3	(k) 76.1	(l) 83.0	(m) 75.0	(n) 81.8	(o) 5.2	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	134.7	80.5	74.2	129.0	79.9	72.8	122.7	79.0	41.1	84.1	39.9	83.1	38.8	81.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.6	(b) 12.3	(c) 10.6	(e) 9.7	(f) 93.8	(g) 4.4	(h) 2.2	(i) 6.5	(j) 95.4	(k) 3.9	(l) 96.7	(m) 1.4	(n) 97.9	(o) -1.8	(p) 99.5		
			(e) 6.8	(f) 78.8	(g) 4.1	(h) 1.6	(i) 3.8	(j) 79.9	(k) 1.5	(l) 80.8	(m) -0.8	(n) 81.7	(o) -3.8	(p) 82.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	32.1	35.0	43.3	53.5	63.6	68.5	75.5	76.0	67.8	59.1	47.7	36.1
	DBStd	16.50	6.63	6.46	7.89	7.71	6.98	5.25	6.03	5.07	3.97	5.25	7.22	7.38
	HDD50	1832	554	420	242	48	1	0	0	0	0	6	128	433
	HDD65	4641	1019	839	673	353	113	24	1	0	16	189	519	895
	CDD50	3648	0	1	35	154	422	556	791	805	534	289	59	2
	CDD65	984	0	0	1	8	70	130	327	340	100	8	0	0
	CDH74	6836	0	0	2	47	482	721	2730	2583	259	12	0	0
	CDH80	2029	0	0	0	6	86	149	978	784	26	0	0	0
Wind	WSAvg	4.4	5.1	4.6	5.0	5.3	4.7	3.5	3.6	3.4	3.4	4.0	4.8	5.2
Precipitation	PrecAvg	55.2	2.0	2.1	2.8	3.1	3.1	4.4	9.1	10.9	8.7	4.2	3.1	1.5
	PrecMax	82.5	9.8	7.7	6.9	9.8	10.9	13.5	29.8	44.8	22.0	13.7	13.7	9.2
	PrecMin	36.4	0.0	0.0	0.3	0.0	0.1	0.6	2.3	2.2	0.0	0.1	0.0	0.0
	PrecStd	11.0	2.0	1.9	1.7	2.2	2.2	2.8	5.3	7.0	5.9	3.5	3.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	56.6	71.2	78.3	86.6	87.3	92.3	92.8	83.1	75.8	67.7	57.1
		MCWB	40.0	44.7	53.5	56.0	61.9	66.1	74.1	76.6	69.8	59.2	54.8	46.5
	2%	DB	47.5	51.1	65.5	74.5	82.2	84.1	90.3	90.1	79.8	72.2	64.2	52.9
		MCWB	37.9	40.8	50.7	54.4	59.6	65.1	74.3	75.1	68.8	59.6	52.4	44.4
	5%	DB	44.1	47.5	61.2	71.0	79.2	81.1	88.1	87.5	76.7	69.6	61.1	49.9
		MCWB	35.5	38.4	47.2	52.8	59.3	64.8	74.1	74.9	67.5	58.8	51.7	41.5
	10%	DB	41.7	44.8	57.2	66.7	76.3	78.4	85.7	84.3	74.6	67.5	58.4	46.7
		MCWB	34.0	37.1	44.3	50.7	57.8	64.8	73.4	73.8	67.0	58.1	50.1	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	46.9	56.4	59.2	67.5	71.5	78.5	79.0	73.8	67.7	58.9	49.9
		MCDB	48.2	54.9	69.2	73.3	75.4	79.0	86.3	86.6	78.9	69.5	64.2	54.5
	2%	WB	39.6	42.2	51.7	56.9	63.5	69.8	77.2	77.7	71.5	64.2	56.1	45.3
		MCDB	44.6	48.2	64.0	71.1	76.3	77.2	85.0	84.4	75.6	68.1	61.3	51.8
	5%	WB	37.4	40.1	48.5	54.7	61.7	68.5	76.0	76.6	70.0	62.2	53.7	42.3
		MCDB	42.3	45.9	58.7	66.9	73.5	75.5	84.2	82.9	74.0	66.4	58.3	48.1
	10%	WB	35.1	38.1	45.3	52.6	60.2	67.4	74.6	75.6	68.5	60.2	51.1	39.9
		MCDB	40.6	43.6	55.4	62.4	70.9	74.1	83.0	81.6	73.0	65.2	57.0	46.0
Mean Daily Temperature Range	5% DB	MDBR	13.4	12.6	13.4	14.7	14.9	11.6	9.9	10.1	11.1	13.0	12.7	13.1
		MCDBR	14.4	16.1	18.8	20.1	19.4	17.0	13.7	13.9	13.7	15.2	14.3	14.4
		MCWBR	9.9	10.7	10.7	10.1	8.9	7.0	5.6	5.9	7.9	9.3	9.5	10.9
	5% WB	MCDBR	12.0	15.2	17.3	18.4	16.4	12.6	11.7	11.6	10.4	11.4	11.7	12.8
		MCWBR	9.3	11.0	10.6	10.2	9.0	6.1	5.6	6.0	6.8	8.6	9.2	10.6
Clear-Sky Solar Irradiance	taub	0.419	0.484	0.581	0.624	0.629	0.679	0.640	0.568	0.495	0.488	0.452	0.410	
	taud	1.964	1.838	1.680	1.647	1.680	1.642	1.783	1.960	2.085	1.973	1.994	2.018	
	Ebn at Noon	226	226	216	218	220	209	217	229	239	223	213	219	
	Edh at Noon	43	56	72	78	77	80	69	56	47	48	42	38	
All-Sky Solar Radiation	RadAvg	764	1010	1304	1556	1726	1637	1360	1354	1238	1081	765	682	
	RadStd	66	97	129	123	169	136	191	170	149	93	100	58	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	-1.58	N/A	N/A	N/A	N/A	N/A	+173	+92

Nomenclature: See separate page